



VISION SUCCESS · UNA, HIMACHAL PRADESH

FREE STUDY MATERIAL · NO. 01

DIGITAL SAT COMMAND CARD

Every formula the Math section can ask you, the Desmos moves that turn 40-second questions into 8-second ones, and the Reading & Writing traps the College Board reuses. One card. Print it. Keep it above your desk.



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KNOW THE MACHINE

The Exam, In Six Lines

Format	2 sections · 98 questions · 2 hr 14 min · on a laptop
Reading & Writing	54 questions · 64 min (two 32-min modules)
Math	44 questions · 70 min (two 35-min modules)
Scoring	400-1600 · NO negative marking
Adaptive	Module 1 performance sets Module 2 difficulty
Calculator	Built-in Desmos – allowed for ALL of Math

MATH

The Formulae Worth Memorising

SLOPE OF A LINE

$$m = (y_2 - y_1) / (x_2 - x_1)$$

LINE, SLOPE-INTERCEPT

$$y = mx + b$$

QUADRATIC FORMULA

$$x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$$

DISCRIMINANT

$$D = b^2 - 4ac$$

VERTEX OF PARABOLA

$$x = -b / 2a$$

CIRCLE (CENTRE H,K)

$$(x-h)^2 + (y-k)^2 = r^2$$

DISTANCE

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

MIDPOINT

$$((x_1 + x_2) / 2, (y_1 + y_2) / 2)$$

PYTHAGORAS

$$a^2 + b^2 = c^2$$

45-45-90 TRIANGLE

$$\text{sides } x : x : x\sqrt{2}$$

30-60-90 TRIANGLE

$$\text{sides } x : x\sqrt{3} : 2x$$

PERCENT CHANGE

$$(\text{new} - \text{old}) / \text{old} \times 100$$

EXPONENTS

$$a^m \cdot a^n = a^{(m+n)}$$

AVERAGES

$$\text{mean} = \text{sum} / \text{count}$$

CIRCLE AREA / CIRCUM.

$$A = \pi r^2, C = 2\pi r$$

TRIG (RIGHT TRIANGLE)

$$\sin = O/H \quad \cos = A/H \quad \tan = O/A$$

THE UNFAIR ADVANTAGE

Desmos: Do Less Maths

- **Systems of equations – graph, don't solve.**

Type both equations in. The intersection point IS the answer. A question designed to take 40 seconds of substitution takes about 8.

- **Quadratics – read the roots off the curve.**

Graph it and Desmos marks the x-intercepts, the vertex and the y-intercept for you. Zeros, minimum, maximum, axis of symmetry – all visible, none calculated.

- **'For what value of k...' – use a slider.**

Type the equation with k in it; Desmos offers to add a slider. Drag it until the graph does what the question describes, and read k directly.

- **Statistics – let it compute.**

Enter data in a table and Desmos gives mean, median and line of best fit without a single hand calculation.

PRACTISE THIS BEFORE TEST DAY

Desmos is only a weapon if it is a reflex. Use it on every practice set from week one, not for the first time in the exam hall.

READING & WRITING

The Traps They Reuse

- **Extreme words are usually wrong.**

'always', 'never', 'completely', 'impossible', 'proves' – the SAT rewards careful, measured claims. Stuck between two options? Cut the extreme one.

- **Half-right is all wrong.**

The classic trap answer is true for the first half and false for the second. Read every option to its final word before choosing.

- **Answer from the text, not your knowledge.**

If it is not in the passage, it is not the answer – however true it may be in real life.

YOU HAVE THE MAP

Now Get The Guide.

This sheet is the free part. The rest – weekly tests, doubt-clearing the day a doubt appears, mock analysis, and a plan built around your calendar – is what we do in the classroom, in batches of never more than fifteen.

For the SAT specifically: full-length adaptive mocks on real Bluebook-style conditions, plus college shortlisting and scholarship guidance.

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